

# Literature Review — Executive Summary

Frozen-VLM Vision-Language-Action (VLA) Project | Position for Paper 1 | Prepared  
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**VERDICT:** The original Paper-1 hypothesis — "a frozen, non-robot-pretrained VLM + diffusion/flow action head needs no backbone robot-data pretraining" — is already published (SmolVLA, EF-VLA, ReMem-VLA). The architecture is not novel. A narrower, still-open question survives: why does freezing work in some reports (SmolVLA/EF-VLA, high SR) and fail in others (Knowledge Insulation, ~0% SR)? That contradiction is unresolved and is the recommended new position for Paper 1.

## 1. The Hypothesis Under Review

Paper 1's hypothesis, as originally framed: to turn a VLM into a VLA via a Diffusion-Policy-style method, large-scale robot-data pretraining of the backbone is unnecessary. A frozen, web-pretrained VLM already emits vision-language features that are "good enough"; the diffusion/flow-matching decoder (with AdaLN pulling global scene+state context together with the VLM vectors) does the heavy lifting. This review asks two things: (a) has this already been shown, and (b) why do VLA papers finetune the backbone at all — what does that tell us about where the real difficulty lives?

## 2. The Field Has Already Split Into Two Camps

A focused search of 2025–2026 VLA literature shows the field is mid-argument over exactly this question. Two camps disagree, using near-identical architectures to reach opposite conclusions.

| CAMP A — Frozen backbone works                                                                                                                                                                                          | CAMP B — Freezing fails                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SmolVLA</b> (HuggingFace, Jun 2025) — frozen VLM, flow-matching action expert trained from scratch, cross-attn + self-attn, state as single token, community data only. <i>Nearly identical to our architecture.</i> | <b>Knowledge Insulation</b> (Physical Intelligence, May 2025) — states frozen backbone "does not have sufficient representations for robotics—freezing doesn't work," reports ~0% SR frozen. Proposes stop-gradient "insulation" instead of full freeze or full finetune.          |
| <b>EF-VLA</b> (ICLR 2025) — frozen CLIP, early fusion, kept frozen specifically to preserve generalization. 85% SR on unseen goal descriptions, +20% on compositional tasks.                                            | <b>VLM4VLA</b> (Jan 2026) — ablates 24 VLMs. Freezing the <b>vision</b> encoder is catastrophic (−28 to −40 pts SimplerEnv). Freezing word embeddings is nearly free. General VLM capability does <b>not</b> predict control performance; embodied-task finetuning actively hurts. |
| <b>ReMem-VLA</b> — frozen VLM + learnable action queries + action diffusion, explicitly "without large-scale pre-training."                                                                                             |                                                                                                                                                                                                                                                                                    |

### 2.1 Why This Sinks the Original Framing

- The architecture (frozen VLM → cross-attention → flow/diffusion action head, state as a conditioning token, no backbone robot pretraining) is SmolVLA's architecture, published two months before this session's review. We cannot claim the design as a contribution.

- The "language grounding transfers from a frozen backbone" result is not novel either — EF-VLA published it at ICLR 2025 with stronger numbers (85% unseen-goal SR) than our current Language Table results (46–74% on held-out verbs, pending the reconciliation noted in §5).
- "You don't need robot pretraining" is already Camp A's accepted position, contested by Camp B. A third voice restating either side, at capstone compute scale, adds little against Physical Intelligence / HuggingFace / DeepMind-scale efforts.

### 3. Why VLA Papers Finetune the Backbone — the Real Justifications

Four distinct reasons recur in the literature for adapting the VLM on robot data. Distinguishing them matters: only one is neutralized by a diffusion/flow action head (our design choice); the others remain open questions our design does not automatically answer.

| Reason                                                                                                                                                                                                     | Neutralized by a flow/diffusion head?                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(1) Action-output formatting.</b> RT-2 represents actions as discretized text tokens and must finetune the LM head to emit them — the single largest driver of backbone finetuning in the RT-2 lineage. | <b>YES — decisively.</b> A diffusion/flow head consumes VLM features; the VLM never has to emit actions. This is the strongest structural argument our design has.                               |
| <b>(2) Embodiment / camera domain gap.</b> Web images differ from robot camera views (gripper, tabletop, viewpoint). Finetuning adapts visual features to this domain.                                     | <b>NO — only partially.</b> This is VLM4VLA's core finding: freezing the vision encoder is catastrophic. This is the live threat to our claim and must be confronted directly, not assumed away. |
| <b>(3) Catastrophic forgetting / preserve web knowledge</b> during robot co-training.                                                                                                                      | YES, trivially — a frozen backbone cannot forget.                                                                                                                                                |
| <b>(4) Cross-embodiment / dexterous scale</b> ( $\pi 0$ , OpenVLA on Open X-Embodiment) — adapting to many robot bodies and high-frequency dexterous control.                                              | Out of scope — concede this boundary explicitly rather than overclaim.                                                                                                                           |

**Closest cousin and biggest threat:**  $\pi 0$  (Physical Intelligence) pairs a VLM (PaliGemma) with a flow-matching action expert — architecturally close to ours — yet still co-finetunes the backbone. Our entire differentiation from  $\pi 0$  is the word *frozen*. Any version of Paper 1 must explicitly answer: *why can we freeze where  $\pi 0$  chose not to?* The honest answer is scope —  $\pi 0$  targets high-frequency dexterous cross-embodiment control requiring tight visual adaptation; our tasks target semantic/language grounding, where web pretraining plausibly already suffices. This boundary is a limitation to own, not a gap to hide.

**Reference point, not a threat:** Diffusion Policy (Chi et al., RSS 2023) shows a conditional diffusion decoder on plain visual features already solves image→action control with no VLM and no language. This sets the floor: any VLA claim must be measured on the **language** axis, not the control axis — control alone was solved without a VLM.

## 4. Novelty Map — What's Taken vs What's Open

| Claim                                                                                                                                                                              | Status          | Source             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------|
| Frozen VLM + diffusion/flow head, no backbone robot-pretraining works                                                                                                              | TAKEN           | SmolVLA, ReMem-VLA |
| Frozen backbone preserves language generalization to unseen goals                                                                                                                  | TAKEN           | EF-VLA (ICLR 2025) |
| Freezing the vision encoder is catastrophic; freezing language/word embeddings is nearly free                                                                                      | TAKEN (partial) | VLM4VLA            |
| General VLM benchmark capability does not predict downstream control performance                                                                                                   | TAKEN           | VLM4VLA            |
| Reconciling <i>why</i> Camp A and Camp B reach opposite SR outcomes on near-identical frozen-backbone designs, using mechanistic tooling (not just benchmark SR)                   | OPEN            | — no paper found   |
| Isolating AdaLN / global-context conditioning specifically (vs. cross-attention generally) as the mechanism that lets a frozen backbone's language signal reach the action decoder | OPEN            | — no paper found   |

## 5. Recommended Position for Paper 1

**Reframed thesis:** "When and why does a frozen VLM suffice for a diffusion-based VLA?" A mechanistic account showing the language pathway survives backbone freezing while the visual pathway is the true bottleneck — reconciling the frozen-works (SmolVLA/EF-VLA) vs. frozen-fails (Knowledge Insulation) contradiction, and localizing the mechanism (AdaLN global-context conditioning vs. cross-attention) by which the surviving language signal reaches the action decoder.

### 5.1 Why This Position Is Defensible

- It does not compete with SmolVLA/EF-VLA on architecture novelty or benchmark SR — it explains *their disagreement*, which neither paper attempts.
- It is analysis-shaped, not scale-shaped: viable on single-GPU capstone compute, unlike a SOTA-chasing system paper competing with Physical-Intelligence-scale resourcing.
- It reuses interpretability tooling already built this project (LoRA-as-projection, cross-attention load-bearing ablation, anti-random rollout proof, language-causal action-divergence probe) — no new infrastructure required to start.
- It directly engages the one concrete gap VLM4VLA leaves open: it isolates the vision/language freeze dichotomy at the level of components (which layer, which pathway) rather than only at the level of aggregate benchmark SR.

### 5.2 What Must Be Shown to Land This Claim

- **AdaLN isolation (highest priority, currently unverified).** The project's named mechanism is "AdaLN pulls global context." Existing ablations only isolated cross-attention (removing it: +261% loss) — AdaLN itself has not been cleanly turned on/off. Run: AdaLN conditioning ON vs. OFF (zeroed/fixed global context), decoder otherwise identical.
- **Vision-vs-language freeze cross-check on our own setup.** Replicate VLM4VLA's core ablation (freeze vision encoder only vs. freeze language pathway only vs. freeze both) on our architecture and task suite, to confirm or complicate their vision-bottleneck finding under a diffusion/flow head specifically (their ablation used autoregressive-token VLAs, not flow/diffusion — this gap is itself worth stating explicitly in related work).
- **Backbone-sufficiency control.** Frozen VLM vs. frozen random-init encoder, same decoder. If random features collapse and VLM features work, the claim that decoder capacity alone is not doing all the work is bounded and defensible.
- **Explicit related-work reconciliation section.** Cite SmolVLA, EF-VLA, ReMem-VLA (Camp A) and Knowledge Insulation, VLM4VLA (Camp B) directly; state the contradiction in their reported numbers (0% vs. 85% SR under "frozen"); position this paper's contribution as resolving it mechanistically rather than re-litigating which camp is right by SR alone.

### 5.3 Risks to Manage

- **Scope creep back to the old framing.** Any writing that reads as "we show frozen VLMs work" will read as redundant to a reviewer who knows SmolVLA/EF-VLA. Every section should be phrased as "why/when," not "whether."
- **The  $\pi 0$  boundary must be stated, not implied.** Reviewers familiar with  $\pi 0$  will ask why we freeze where they didn't; pre-empt this in related work (dexterous cross-embodiment control vs. semantic/language grounding — see §3).
- **Venue calibration.** This framing is workshop-realistic (CoRL/IROS workshop) on current evidence; a main-venue submission needs the AdaLN and vision/language freeze results in hand before the claim can be judged strong enough to attempt it.

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Sources reviewed: SmolVLA (arXiv 2506.01844), EF-VLA (ICLR 2025, OpenReview KBShR4h8XV), ReMem-VLA (arXiv 2603.12942), Knowledge Insulation (arXiv 2505.23705), VLM4VLA (arXiv 2601.03309),  $\pi 0$  (arXiv 2410.24164), RT-2 (Zitkovich et al., CoRL 2023), Diffusion Policy (Chi et al., RSS 2023). Paper 2 (data-pipeline thesis) intentionally out of scope for this summary per current focus.